

# ARUL JEFFRY A

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## CAREER OBJECTIVE

Final-year Electrical and Electronics Engineering undergraduate (2027 batch, SASTRA Deemed University) with hands-on experience in power electronics, power conversion, closed-loop control, multi-layer PCB design using KiCad, and embedded firmware development on TI C2000 microcontrollers. Seeking a Graduate Engineer Trainee (GET) / Project Engineer role to contribute to power conversion system design, circuit design, hardware debugging, and embedded control for industrial and electrification applications.

## EDUCATION

**SASTRA Deemed to be University, Thanjavur**

2023 - 2027

B.Tech, Electrical and Electronics Engineering (Smart Grid and Electric Vehicles) | CGPA: 7.115/10 | No standing arrears

**Higher Secondary Certificate (HSC)**

St. Joseph's College Hr Sec School | 80%

## TECHNICAL SKILLS

- **Power Electronics:** Buck/SIMO DC-DC Converters, PWM Generation, PI Closed-Loop Control, Power Conversion System Design
- **Microcontrollers & Embedded:** TI C2000 (F280049C, F28069), STM32F103C8T6, Embedded C, ADC/DAC, ePWM, I<sup>2</sup>C, SPI, Peripheral Interfacing
- **PCB Design & Fabrication:** KiCad, Schematic Capture, Multi-layer PCB Layout, Power/Ground Planes, DRC, Gerber/BOM Generation, PCB Fabrication & Soldering
- **Power Systems:** Energy Metering, Protection Equipment (CTs, PTs, Relays, Circuit Breakers), Substation & SCADA Monitoring
- **Development Tools:** MATLAB/Simulink, Code Composer Studio, STM32CubeIDE
- **Programming Languages:** C, C++, Python

## INTERNSHIP EXPERIENCE

**TANGEDCO (Tamil Nadu Generation and Distribution Corporation Limited), Trichy Electricity Distribution Circle**

Electrical Engineering Intern

Dec 2025

- **Energy Metering & Testing:** Studied energy meter types, operating principles, and testing procedures.
- **Protection & Power Equipment:** Covered CTs, PTs, transformers, relays, and circuit breakers, including their operating principles and ratings.
- **Substation & SCADA:** Analyzed substation single-line diagrams and SCADA-based voltage and current monitoring.

## PROJECTS

**SIMO DC-DC Buck Converter – Design & Simulation** MATLAB-SIMULINK May 2026

Design Calculations Lead | Three-member team | SASTRA Deemed to be University

- Designed a six-switch, five-inductor SIMO buck converter generating regulated 18.5V, 15V, 12V, 5V, and 3.3V outputs from a 48V DC input.
- Performed duty-cycle, inductance, and capacitance calculations for Continuous Conduction Mode (CCM) with a 0.02% output-ripple target.
- Developed the MATLAB/Simulink model with PI closed-loop control and NAND-based switching logic and analyzed voltage, current, PWM, and ripple waveforms.

**TMS320F28069 Synchronous Buck Converter – Embedded Firmware** Code Composer Studio

- Developed firmware for a TI TMS320F28069-based synchronous buck converter, implementing complementary PWM generation, deadband, and ePWM configuration.

**LAUNCHXL-F280049C – Four-Layer PCB Design** KiCad

- Designed a four-layer TI C2000 LaunchPad PCB in KiCad with power/ground planes, component placement, and PCB routing.
- Performed Design Rule Check (DRC) and generated Gerber, drill, and BOM files for PCB fabrication.

**STM32F103C8T6 – Two-Layer PCB Design** KiCad

- Designed a two-layer STM32F103C8T6 PCB with schematic capture, ground pour, and PCB routing.
- Performed Design Rule Check (DRC) and generated Gerber files for PCB fabrication.

## CERTIFICATIONS

- PCB Design Course – MHI Training Centre, SASTRA Deemed to be University
- Introduction to MATLAB and Simulink Workshop